

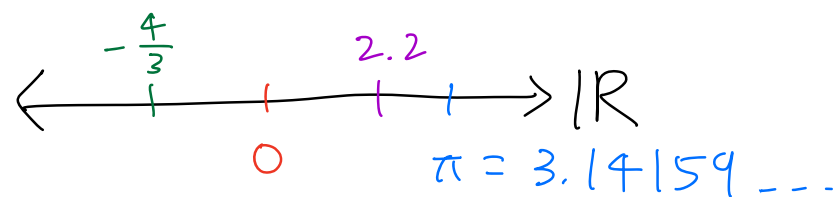
Unit 1: Vector Spaces

Goal: Define vectors & vector spaces, then study their properties (so build foundations)

§1.1 $\mathbb{R}^n$ as a Vector Space

Before jumping into $\mathbb{R}^n$, let's start with $\mathbb{R}^2$.

Here, $\mathbb{R}$ means the set of all real numbers

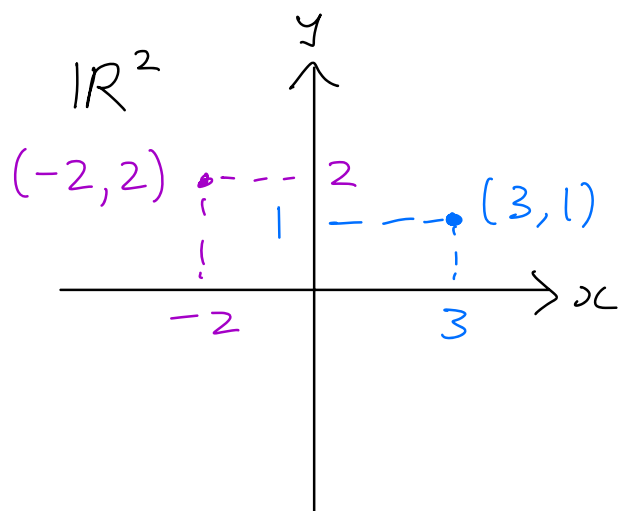


Definition

Def 1.1: $\mathbb{R}^2 \stackrel{\text{LHS}}{:=} \{ \underbrace{(a, b)}_{\text{a pair}} : \underbrace{a, b \in \mathbb{R}}_{\text{means } a \text{ \& } b \text{ are elements of } \mathbb{R}} \}$
(Left-hand side is defined by Right-hand side)

Note: So $\mathbb{R}^2$ is a set of pairs of real numbers.
2-tuples

Cartesian plane is a way to visualize $\mathbb{R}^2$

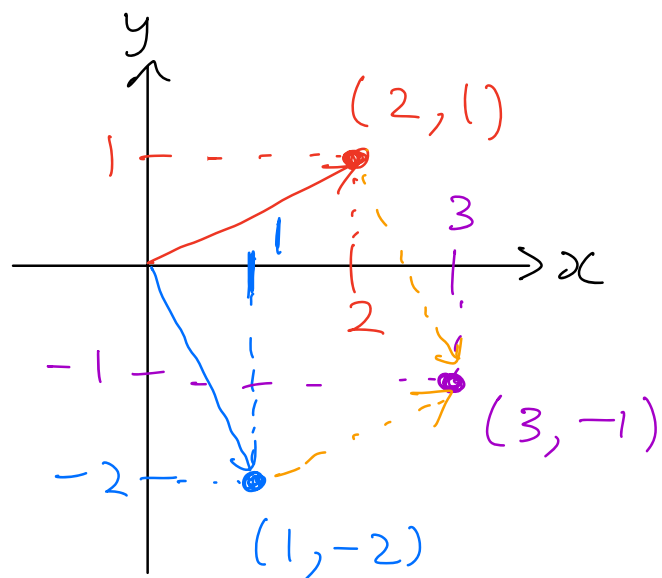


There are two operations on $\mathbb{R}^2$:

(1) Coordinate-wise Addition: If $(a_1, a_2), (b_1, b_2) \in \mathbb{R}^2$,
then $(a_1, a_2) + (b_1, b_2) := (a_1 + b_1, a_2 + b_2) \in \mathbb{R}^2$

For example,

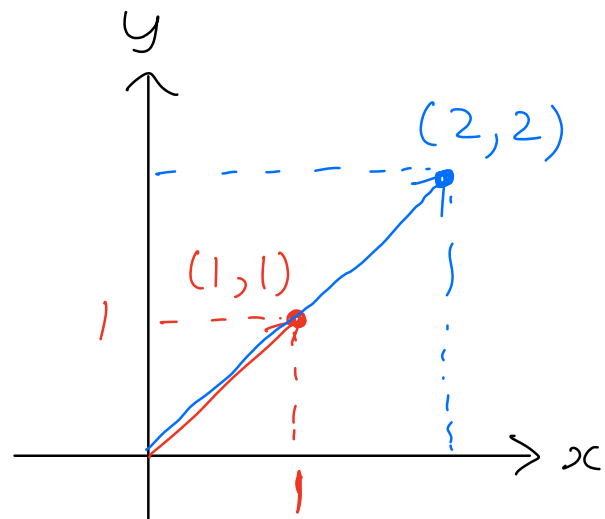
$$\begin{aligned} (2, 1) + (1, -2) &= (2+1, 1-2) \\ &= (3, -1) \end{aligned}$$



(2) Scalar - Multiplication: If $c \in \mathbb{R}$ and $(x, y) \in \mathbb{R}^2$,
then $c \cdot (x, y) := (cx, cy)$.

For example,

$$2 \cdot (1, 1) = (2 \cdot 1, 2 \cdot 1) = (2, 2)$$

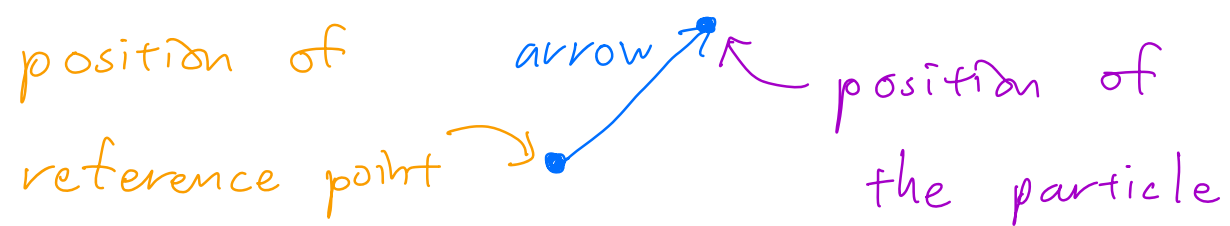


Note: Subtraction on $\mathbb{R}^2$ is

$$\begin{aligned} (a, b) - (c, d) &:= (a, b) + (-1) \cdot (c, d) = (a, b) + (-c, -d) \\ &= (a - c, b - d) \end{aligned}$$

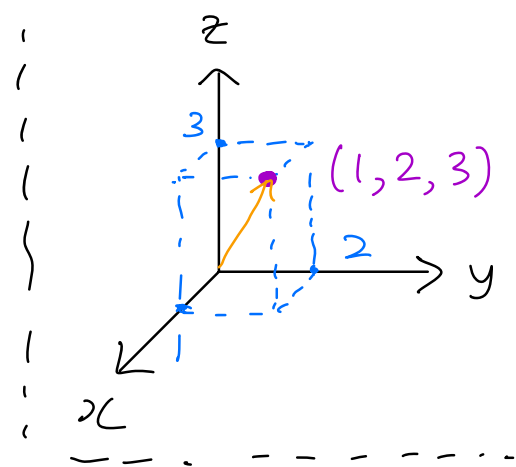
Rmk: In Physics, displacement (relative position

of a particle from fixed reference point) on a plane can be modelled as a point of $\mathbb{R}^2$



Similarly, $\mathbb{R}^3$ has many properties similar to $\mathbb{R}^2$:

Def 1.2: $\mathbb{R}^3 := \{(a, b, c) : a, b, c \in \mathbb{R}\}$
3-tuple of real numbers



$\mathbb{R}^3$ has both $+$ & $\cdot$:

$$(a_1, a_2, a_3) + (b_1, b_2, b_3) := (a_1 + b_1, a_2 + b_2, a_3 + b_3)$$

$$c \cdot (a_1, a_2, a_3) := (ca_1, ca_2, ca_3)$$

whenever $a_1, a_2, a_3, b_1, b_2, b_3, c \in \mathbb{R}$.

$$\left(\begin{aligned} &\text{and } (a_1, a_2, a_3) - (b_1, b_2, b_3) \\ &= (a_1, a_2, a_3) + (-1) \cdot (b_1, b_2, b_3) \\ &= (a_1 - b_1, a_2 - b_2, a_3 - b_3) \end{aligned} \right)$$

Mathematician's

Guiding Principle: when people say a "vector", they

mean "things" that can be added & scalar multiplied. Similarly, "vector space" means a set of "vectors" (with additional properties). In this sense, both $\mathbb{R}^2$ & $\mathbb{R}^3$ are vector spaces. (We will precisely define these words in Section 1.2 §1.2)

$\mathbb{R}^2$ represents 2-dimensional plane and $\mathbb{R}^3$ represents 3-dimensional space. So to represent higher-dimensional space (e.g. spacetime), need to generalize $\mathbb{R}^2$ & $\mathbb{R}^3$ into $\mathbb{R}^n$:

Def 1.3: Fix $n \in \mathbb{Z}_{>0}$ ^{set of positive integers} $:= \{1, 2, 3, \dots\}$. Then

$\mathbb{R}^n := \{(\underbrace{x_1, \dots, x_n}_{n\text{-tuple of real numbers}}) : x_i \in \mathbb{R} \text{ for every } i=1, \dots, n\}$.

n -tuple of real numbers

Given $(x_1, \dots, x_n) \in \mathbb{R}^n$, its $\underbrace{k^{\text{th}}}_{\substack{k \in \{1, \dots, n\} \\ \text{coord}}}$ -coordinate is x_k .

Rmk: Some books def $\mathbb{R}^0 := \{0\}$.

Def 1.4: Addition and scalar-multiplication ^{mult} on

$\mathbb{R}^n$ is:

$$(1) \forall (x_1, \dots, x_n), (y_1, \dots, y_n) \in \mathbb{R}^n,$$

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) := (x_1 + y_1, \dots, x_n + y_n) \in \mathbb{R}^n$$

$$(2) \forall (x_1, \dots, x_n) \in \mathbb{R}^n \text{ \& \> } \forall c \in \mathbb{R},$$

$$c \cdot (x_1, \dots, x_n) = c(x_1, \dots, x_n) := (cx_1, \dots, cx_n)$$

To extract properties of $\mathbb{R}^n$, we need to re-interpret $+$ & $\cdot$ via sets and functions (also called maps):

Def: (Cartesian) product of two sets X, Y is

$$X \times Y := \{(x, y) : x \in X, y \in Y\}.$$

$f: X \rightarrow Y$ function
 $x \mapsto f(x)$ x maps to $f(x)$

Example: $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$, $\mathbb{R}^3 \xleftarrow[\text{(-1) con}]{\text{(-1) onto}} \mathbb{R}^2 \times \mathbb{R}$

$$(a, b, c) \mapsto ((a, b), c)$$

Note 1.5: Def 1.4 & Prev Def

$\Rightarrow +$ & $\cdot$ are ^{funs} functions as well:

$$+ : \mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

$$((x_1, \dots, x_n), (y_1, \dots, y_n)) \mapsto (x_1 + y_1, \dots, x_n + y_n)$$

$$\cdot : \mathbb{R} \times \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

$$(c, (x_1, \dots, x_n)) \mapsto (cx_1, \dots, cx_n)$$

↓

Proposition

Prop 1.6: $\mathbb{R}^n$ with $+$ (vector addition) and $\cdot$ (scalar multiplication) satisfying:

$\cdot : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$

$$(VS 1) \forall v, w \in \mathbb{R}^n, v + w = w + v$$

$$(VS 2) \forall u, v, w \in \mathbb{R}^n, (u + v) + w = u + (v + w)$$

(VS 3) There $\exists$ exists $0_{\mathbb{R}^n} \in \mathbb{R}^n$ (or simply called 0) such that $\forall v \in \mathbb{R}^n, v + 0_{\mathbb{R}^n} = v = 0_{\mathbb{R}^n} + v$

$$(VS 4) \forall v \in \mathbb{R}^n, \exists w \in \mathbb{R}^n \text{ s.t. } v + w = 0_{\mathbb{R}^n}$$

$$(VS 5) \forall v \in \mathbb{R}^n, 1 \cdot v = v$$

$$(VS 6) \forall a, b \in \mathbb{R} \text{ and } \forall v \in \mathbb{R}^n, (ab) \cdot v = a \cdot (b \cdot v)$$

$$(VS 7) \forall c \in \mathbb{R} \text{ and } \forall v, w \in \mathbb{R}^n, c(v + w) = cv + cw$$

$$(VS 8) \forall a, b \in \mathbb{R} \text{ and } \forall v \in \mathbb{R}^n, (a + b)v = av + bv$$

Remark

Rmk 1.7:

(a) (VS 1) – (VS 8) tell us that $+$ & $\cdot$ behave “properly” (matches w/ our intuition)

(b) $0_{\mathbb{R}^n} \in \mathbb{R}^n$ from (VS 3) is unique.

$0_{\mathbb{R}^n} = (0, \dots, 0) = \mathbf{0}$ is called a zero vector
abuse of notation

(c) From (VS4), the $w \in \mathbb{R}^n$ is just $(-1)v$.

We often write $-v$ as $(-1)v$. So that w is unique.

Idea
Pf of Prop 1.6: Because $\mathbb{R} = \mathbb{R}^1$ satisfies
(VS1) – (VS8) (Korean high school math) and
 $+$: $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ & $\cdot$: $\mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is def
coordinate-wise, $n=1$ case "implies" $n>1$ case.

this needs full-justification

□

Notation 1.7: In this course, we need to view

$\mathbb{R}^n$ as a "vector space" of "column vectors".

(mathematician's viewpoint)

n rows,
single column,
each $x_i \in \mathbb{R} \ \forall i$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$:= (x_1, \dots, x_n)$$

n -tuple of $\mathbb{R}$ #'s

col vec
column vector of size n (over $\mathbb{R}$)

$\Downarrow$

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_i \in \mathbb{R} \ \forall i \right\} = \{ (x_1, \dots, x_n) : x_i \in \mathbb{R} \ \forall i \}$$

and $\left\{ \begin{array}{l} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{bmatrix} \quad \forall \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}, \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} \in \mathbb{R}, \end{array} \right.$

Def
1.4

$$\left\{ \begin{array}{l} c \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} ca_1 \\ \vdots \\ ca_n \end{bmatrix} \quad \forall \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R} \text{ \& } \forall c \in \mathbb{R} \end{array} \right.$$